TURNING 10 YEARS OF GRID DATA INTO ENGINEERING INSIGHT

Building a searchable database and automated analysis workflow from historical operating data

Mason McGregor | Electrical Engineering Intern | Black Hills Energy | South Dakota State University

Black Hills Energy — Improving life with energy

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1) THE PROBLEM / BACKGROUND

BEFORE: HARD TO USE

- Raw text files
- Hard to search or filter
- Time-consuming manual review
- Inconsistent formatting and naming
- Bad or missing data can distort results

AFTER: EASY TO USE

- Searchable database
- Clean, consistent data
- Automated analysis
- Clear visualizations
- Reliable insights for engineering decisions

2) OBJECTIVE

Create a repeatable workflow that transforms 10 years of historical grid operating data into clean, searchable, and useful engineering insight for planning, modeling, and engineering studies.

4) KEY FEATURES

- Processed ~10 years of hourly data
- Extracted timestamps, MW, MVAR, and metadata
- Standardized component names and source fields
- Cleaned missing, bad, and impossible values
- Automated summary statistics and event views
- Generated repeatable visual reports for engineers

7) CHALLENGES & LESSONS LEARNED

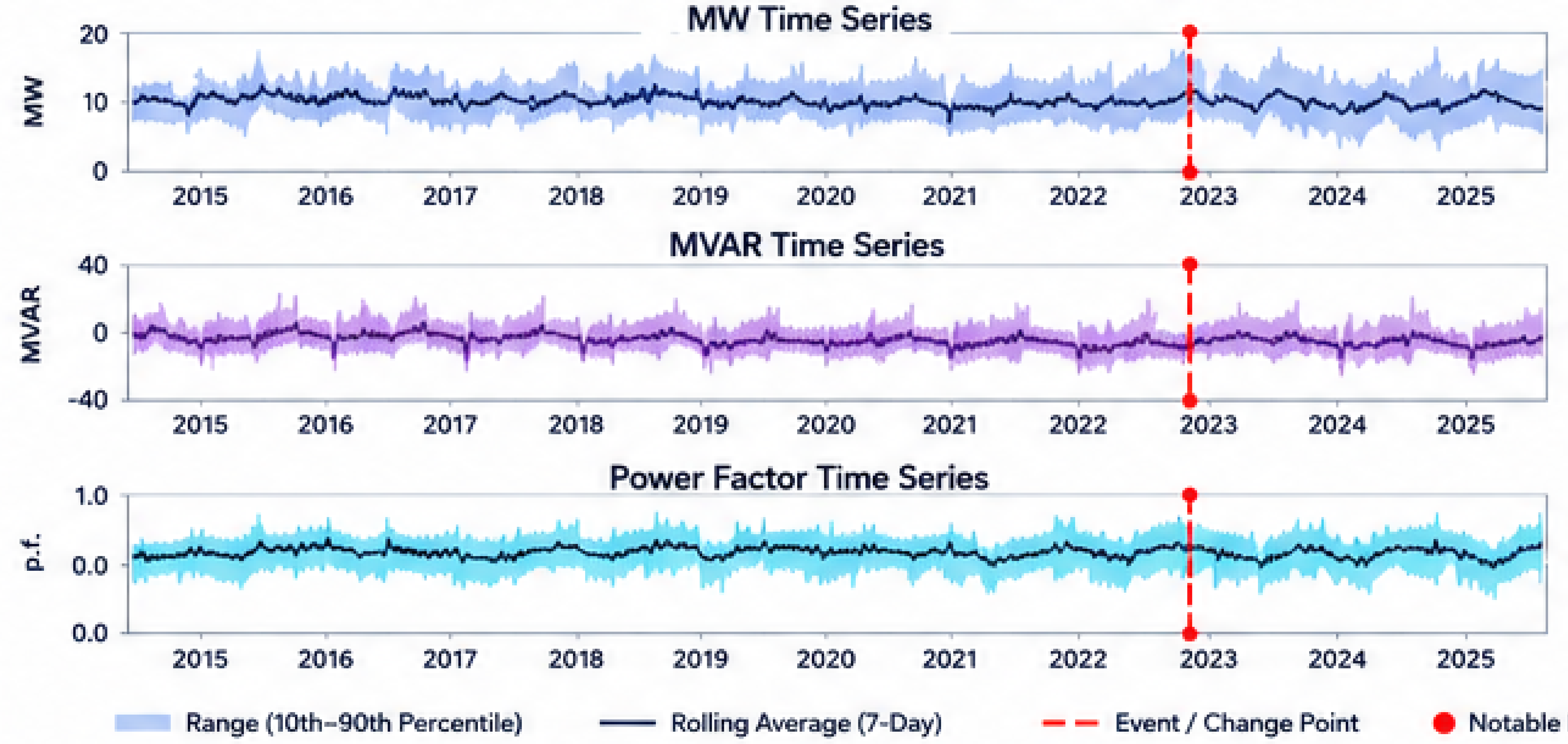
- Standardized inconsistent naming across 10 years of data
- Filtered impossible telemetry values without hiding real events
- Built tools that reduce manual effort for engineers

3) SYSTEM OVERVIEW

1. **Raw SCADA Files** — Hourly operating data from grid equipment
2. **Parsing & Cleaning** — Extract, clean, standardize, and validate data
3. **Searchable Access Database** — Structured indexed database for fast queries and access
4. **Statistical Analysis** — Calculate key metrics, detect events and patterns
5. **Automated Reports & Plots** — Generate repeatable plots and summary reports
6. **Engineering Decisions** — Actionable insight that informs studies and planning

An end-to-end pipeline converts raw historical grid data into clean, organized, and useful information engineers can quickly use.

5) RESULTS / EXAMPLE OUTPUT



Seasonal patterns and long-term trends are clear and consistent.

Outliers and variance spikes are detected automatically.

Operating behavior stays within expected ranges.

6) ENGINEERING IMPACT

Faster Studies
Reduces time to pull seasonal and component-level data.

Better System Models
Supports more accurate and defensible models using real data.

Reduced Planning Risk
Improves confidence in grid behavior and planning decisions.

Compliance Support
Supports case building, load allocation, and UFLS-related analysis.

Reusable Workflow
Creates a repeatable process for future studies and updates.

8) FUTURE WORK

- Expand to additional data sources and components
- Add more automated reporting and dashboards
- Integrate the workflow into future engineering study processes

9) WHAT IS SCADA DATA?

SCADA data is hourly operating data collected from equipment across the electric grid. It includes values such as MW, MVAR, timestamps, component names, and data-quality flags.

10) WORKFLOW BENEFITS

- ✓ Analyze MW, MVAR, and power factor
- ✓ Show seasonal and long-term trends
- ✓ Identify extreme events and outliers
- ✓ Provide summary statistics
- ✓ Find when components started operating
- ✓ Support quick ad hoc engineering reviews

KEY TAKEAWAY: This project transformed 10 years of raw grid operating data into a searchable database and automated analysis workflow—helping engineers make faster, more informed, and more confident decisions.